For the most part in this chapter, we have focused on bilateral $z$-transforms. However, as with the Laplace transform, we have also introduced a second form of the $z$-transform known as the unilateral $z$-transform. The unilateral transform, which can be viewed as the bilateral transform of a signal whose values for $n < 0$ have been set to zero, is particularly useful for analyzing systems described by linear constant-coefficient difference equations with nonzero initial conditions.

# Chapter 10 Problems

The first section of problems belongs to the basic category, and the answers are provided in the back of the book. The remaining three sections contain problems belonging to the basic, advanced, and extension categories, respectively.

## BASIC PROBLEMS WITH ANSWERS

**10.1.** Determine the constraint on $r = |z|$ for each of the following sums to converge:

(a) $\displaystyle\sum_{n=-1}^{\infty} (\tfrac{1}{2})^{n+1} z^{-n}$ (b) $\displaystyle\sum_{n=1}^{\infty} (\tfrac{1}{2})^{-n+1} z^{n}$

(c) $\displaystyle\sum_{n=0}^{\infty} \{\tfrac{1+(-1)^n}{2}\} z^{-n}$ (d) $\displaystyle\sum_{n=-\infty}^{\infty} (\tfrac{1}{2})^{|n|} \cos(\tfrac{\pi}{4} n) z^{-n}$

**10.2.** Consider the signal

$$x[n] = \left(\frac{1}{5}\right)^n u[n-3].$$

Use eq. (10.3) to evaluate the $z$-transform of this signal, and specify the corresponding region of convergence

**10.3.** Let

$$x[n] = (-1)^n u[n] + \alpha^n u[-n-n_0].$$

Determine the constraints on the complex number $\alpha$ and the integer $n_0$, given that the ROC of $X(z)$ is

$$1 < |z| < 2.$$

**10.4.** Consider the signal

$$x[n] = \begin{cases} (\tfrac{1}{3})^n \cos(\tfrac{\pi}{2} n), & n \leq 0 \\ 0, & n > 0 \end{cases}.$$

Determine the poles and ROC for $X(z)$.

**10.5.** For each of the following algebraic expressions for the $z$-transform of a signal, determine the number of zeros in the finite $z$-plane and the number of zeros at infinity.

**(a)** $\dfrac{z^{-1}(1-\frac{1}{2}z^{-1})}{(1-\frac{1}{3}z^{-1})(1-\frac{1}{4}z^{-1})}$

**(b)** $\dfrac{(1-z^{-1})(1-2z^{-1})}{(1-3z^{-1})(1-4z^{-1})}$

**(c)** $\dfrac{z^{-2}(1-z^{-1})}{(1-\frac{1}{4}z^{-1})(1+\frac{1}{4}z^{-1})}$

**10.6.** Let $x[n]$ be an absolutely summable signal with rational $z$-transform $X(z)$. If $X(z)$ is known to have a pole at $z = 1/2$, could $x[n]$ be
**(a)** a finite-duration signal?
**(b)** a left-sided signal?
**(c)** a right-sided signal?
**(d)** a two-sided signal?

**10.7.** Suppose that the algebraic expression for the $z$-transform of $x[n]$ is

$$X(z) = \frac{1-\frac{1}{4}z^{-2}}{(1+\frac{1}{4}z^{-2})(1+\frac{5}{4}z^{-1}-\frac{3}{8}z^{-2})}.$$

How many different regions of convergence could correspond to $X(z)$?

**10.8.** Let $x[n]$ be a signal whose rational $z$-transform $X(z)$ contains a pole at $z = 1/2$. Given that

$$x_1[n] = \left(\frac{1}{4}\right)^n x[n]$$

is absolutely summable and

$$x_2[n] = \left(\frac{1}{8}\right)^n x[n]$$

is not absolutely summable, determine whether $x[n]$ is left sided, right sided, or two sided.

**10.9.** Using partial-fraction expansion and the fact that

$$a^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1-az^{-1}}, \quad |z| > |a|,$$

find the inverse $z$-transform of

$$X(z) = \frac{1-\frac{1}{3}z^{-1}}{(1-z^{-1})(1+2z^{-1})}, \quad |z| > 2.$$

**10.10.** Consider the following algebraic expression for the $z$-transform $X(z)$ of a signal $x[n]$:

$$X(z) = \frac{1+z^{-1}}{1+\frac{1}{3}z^{-1}}.$$

(a) Assuming the ROC to be $|z| > 1/3$, use long division to determine the values of $x[0]$, $x[1]$, and $x[2]$.
(b) Assuming the ROC to be $|z| < 1/3$, use long division to determine the values of $x[0]$, $x[-1]$, and $x[-2]$.

**10.11.** Find the inverse $z$-transform of

$$X(z) = \frac{1}{1{,}024}\left[\frac{1{,}024 - z^{-10}}{1 - \frac{1}{2}z^{-1}}\right], \quad |z| > 0.$$

**10.12.** By considering the geometric interpretation of the magnitude of the Fourier transform from the pole-zero plot, determine, for each of the following $z$-transforms, whether the corresponding signal has an approximately lowpass, bandpass, or highpass characteristic:

(a) $X(z) = \dfrac{z^{-1}}{1 + \frac{8}{9}z^{-1}}, \quad |z| > \frac{8}{9}$

(b) $X(z) = \dfrac{1 + \frac{8}{9}z^{-1}}{1 - \frac{16}{9}z^{-1} + \frac{64}{81}z^{-2}}, \quad |z| > \frac{8}{9}$

(c) $X(z) = \dfrac{1}{1 + \frac{64}{81}z^{-2}}, \quad |z| > \frac{8}{9}$

**10.13.** Consider the rectangular signal

$$x[n] = \begin{cases} 1, & 0 \le n \le 5 \\ 0, & \text{otherwise} \end{cases}.$$

Let

$$g[n] = x[n] - x[n-1].$$

(a) Find the signal $g[n]$ and directly evaluate its $z$-transform.
(b) Noting that

$$x[n] = \sum_{k=-\infty}^{n} g[k],$$

use Table 10.1 to determine the $z$-transform of $X(z)$.

**10.14.** Consider the triangular signal

$$g[n] = \begin{cases} n - 1, & 2 \le n \le 7 \\ 13 - n, & 8 \le n \le 12 \\ 0, & \text{otherwise} \end{cases}.$$

(a) Determine the value of $n_0$ such that

$$g[n] = x[n] * x[n - n_0],$$

where $x[n]$ is the rectangular signal considered in Problem 10.13.
(b) Use the convolution and shift properties in conjunction with $X(z)$ found in Problem 10.13 to determine $G(z)$. Verify that your answer satisfies the initial-value theorem.

**10.15.** Let

$$y[n] = \left(\frac{1}{9}\right)^n u[n].$$

Determine two distinct signals such that each has a $z$-transform $X(z)$ which satisfies both of the following conditions:

1. $[X(z) + X(-z)]/2 = Y(z^2)$.
2. $X(z)$ has only one pole and only one zero in the $z$-plane.

**10.16.** Consider the following system functions for stable LTI systems. Without utilizing the inverse $z$-transform, determine in each case whether or not the corresponding system is causal.

**(a)** $\dfrac{1 - \frac{4}{3}z^{-1} + \frac{1}{2}z^{-2}}{z^{-1}(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}$

**(b)** $\dfrac{z - \frac{1}{2}}{z^2 + \frac{1}{2}z - \frac{3}{16}}$

**(c)** $\dfrac{z + 1}{z + \frac{4}{3} - \frac{1}{2}z^{-2} - \frac{2}{3}z^{-3}}$

**10.17.** Suppose we are given the following five facts about a particular LTI system $S$ with impulse response $h[n]$ and $z$-transform $H(z)$:

1. $h[n]$ is real.
2. $h[n]$ is right sided.
3. $\lim_{z \to \infty} H(z) = 1$.
4. $H(z)$ has two zeros.
5. $H(z)$ has one of its poles at a nonreal location on the circle defined by $|z| = 3/4$.

Answer the following two questions:

**(a)** Is $S$ causal? **(b)** Is $S$ stable?

**10.18.** Consider a causal LTI system whose input $x[n]$ and output $y[n]$ are related through the block diagram representation shown in Figure P10.18.

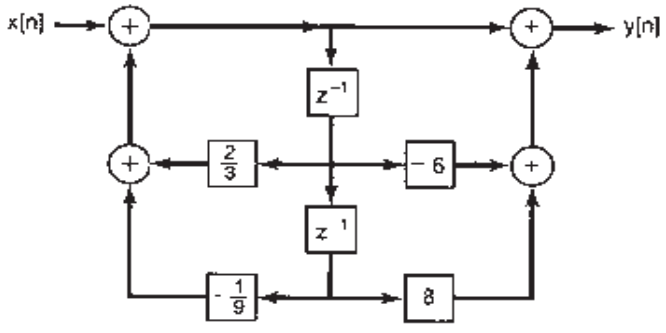



Figure P10.18

**(a)** Determine a difference equation relating $y[n]$ and $x[n]$.

**(b)** Is this system stable?

**10.19.** Determine the unilateral $z$-transform of each of the following signals, and specify the corresponding regions of convergence:

(a) $x_1[n] = (\frac{1}{4})^n u[n+5]$
(b) $x_2[n] = \delta[n+3] + \delta[n] + 2^n u[-n]$
(c) $x_3[n] = (\frac{1}{2})^{|n|}$

**10.20.** Consider a system whose input $x[n]$ and output $y[n]$ are related by

$$y[n-1] + 2y[n] = x[n].$$

(a) Determine the zero-input response of this system if $y[-1] = 2$.
(b) Determine the zero-state response of the system to the input $x[n] = (1/4)^n u[n]$.
(c) Determine the output of the system for $n \geq 0$ when $x[n] = (1/4)^n u[n]$ and $y[-1] = 2$.

## BASIC PROBLEMS

**10.21.** Determine the $z$-transform for each of the following sequences. Sketch the pole-zero plot and indicate the region of convergence. Indicate whether or not the Fourier transform of the sequence exists.

(a) $\delta[n+5]$ (b) $\delta[n-5]$
(c) $(-1)^n u[n]$ (d) $(\frac{1}{2})^{n+1} u[n+3]$
(e) $(-\frac{1}{3})^n u[-n-2]$ (f) $(\frac{1}{4})^n u[3-n]$
(g) $2^n u[-n] + (\frac{1}{4})^n u[n-1]$ (h) $(\frac{1}{3})^{n-2} u[n-2]$

**10.22.** Determine the $z$-transform for the following sequences. Express all sums in closed form. Sketch the pole-zero plot and indicate the region of convergence. Indicate whether the Fourier transform of the sequence exists.

(a) $(\frac{1}{2})^n\{u[n+4] - u[n-5]\}$ (b) $n(\frac{1}{2})^{|n|}$
(c) $|n|(\frac{1}{2})^{|n|}$ (d) $4^n \cos[\frac{2\pi}{6}n + \frac{\pi}{4}]u[-n-1]$

**10.23.** Following are several $z$-transforms. For each one, determine the inverse $z$-transform using both the method based on the partial-fraction expansion and the Taylor's series method based on the use of long division.

$$X(z) = \frac{1 - z^{-1}}{1 - \frac{1}{4}z^{-2}}, \quad |z| > \frac{1}{2}.$$

$$X(z) = \frac{1 - z^{-1}}{1 - \frac{1}{4}z^{-2}}, \quad |z| < \frac{1}{2}.$$

$$X(z) = \frac{z^{-1} - \frac{1}{2}}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}.$$

$$X(z) = \frac{z^{-1} - \frac{1}{2}}{1 - \frac{1}{2}z^{-1}}, \quad |z| < \frac{1}{2}.$$

$$X(z) = \frac{z^{-1} - \frac{1}{2}}{(1 - \frac{1}{2}z^{-1})^2}, \quad |z| > \frac{1}{2}.$$

$$X(z) = \frac{z^{-1} - \frac{1}{2}}{(1 - \frac{1}{2}z^{-1})^2}, \quad |z| < \frac{1}{2}.$$

**10.24.** Using the method indicated, determine the sequence that goes with each of the following $z$-transforms:

**(a)** Partial fractions:

$$X(z) = \frac{1 - 2z^{-1}}{1 - \frac{5}{2}z^{-1} + z^{-2}}, \text{ and } x[n] \text{ is absolutely summable.}$$

**(b)** Long division:

$$X(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{1}{2}z^{-1}}, \text{ and } x[n] \text{ is right sided.}$$

**(c)** Partial fractions:

$$X(z) = \frac{3}{z - \frac{1}{4} - \frac{1}{8}z^{-1}}, \text{ and } x[n] \text{ is absolutely summable.}$$

**10.25.** Consider a right-sided sequence $x[n]$ with $z$-transform

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}. \qquad \text{(P10.25–1)}$$

**(a)** Carry out a partial-fraction expansion of eq. (P10.25–1) expressed as a ratio of polynomials in $z^{-1}$, and from this expansion, determine $x[n]$.

**(b)** Rewrite eq. (P10.25–1) as a ratio of polynomials in $z$, and carry out a partial-fraction expansion of $X(z)$ expressed in terms of polynomials in $z$. From this expansion, determine $x[n]$, and demonstrate that the sequence obtained is identical to that obtained in part (a).

**10.26.** Consider a left-sided sequence $x[n]$ with $z$-transform

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}.$$

**(a)** Write $X(z)$ as a ratio of polynomials in $z$ instead of $z^{-1}$.

**(b)** Using a partial-fraction expression, express $X(z)$ as a sum of terms, where each term represents a pole from your answer in part (a).

**(c)** Determine $x[n]$.

**10.27.** A right-sided sequence $x[n]$ has $z$-transform

$$X(z) = \frac{3z^{-10} + z^{-7} - 5z^{-2} + 4z^{-1} + 1}{z^{-10} - 5z^{-7} + z^{-3}}.$$

Determine $x[n]$ for $n < 0$.

**10.28.** **(a)** Determine the $z$-transform of the sequence

$$x[n] = \delta[n] - 0.95\,\delta[n - 6].$$

**(b)** Sketch the pole-zero pattern for the sequence in part (a).

**(c)** By considering the behavior of the pole and zero vectors as the unit circle is traversed, develop an approximate sketch of the magnitude of the Fourier transform of $x[n]$.

**10.29.** By considering the geometric determination of the frequency response as discussed in Section 10.4, sketch, for each of the pole-zero plots in Figure P10.29, the magnitude of the associated Fourier transform.

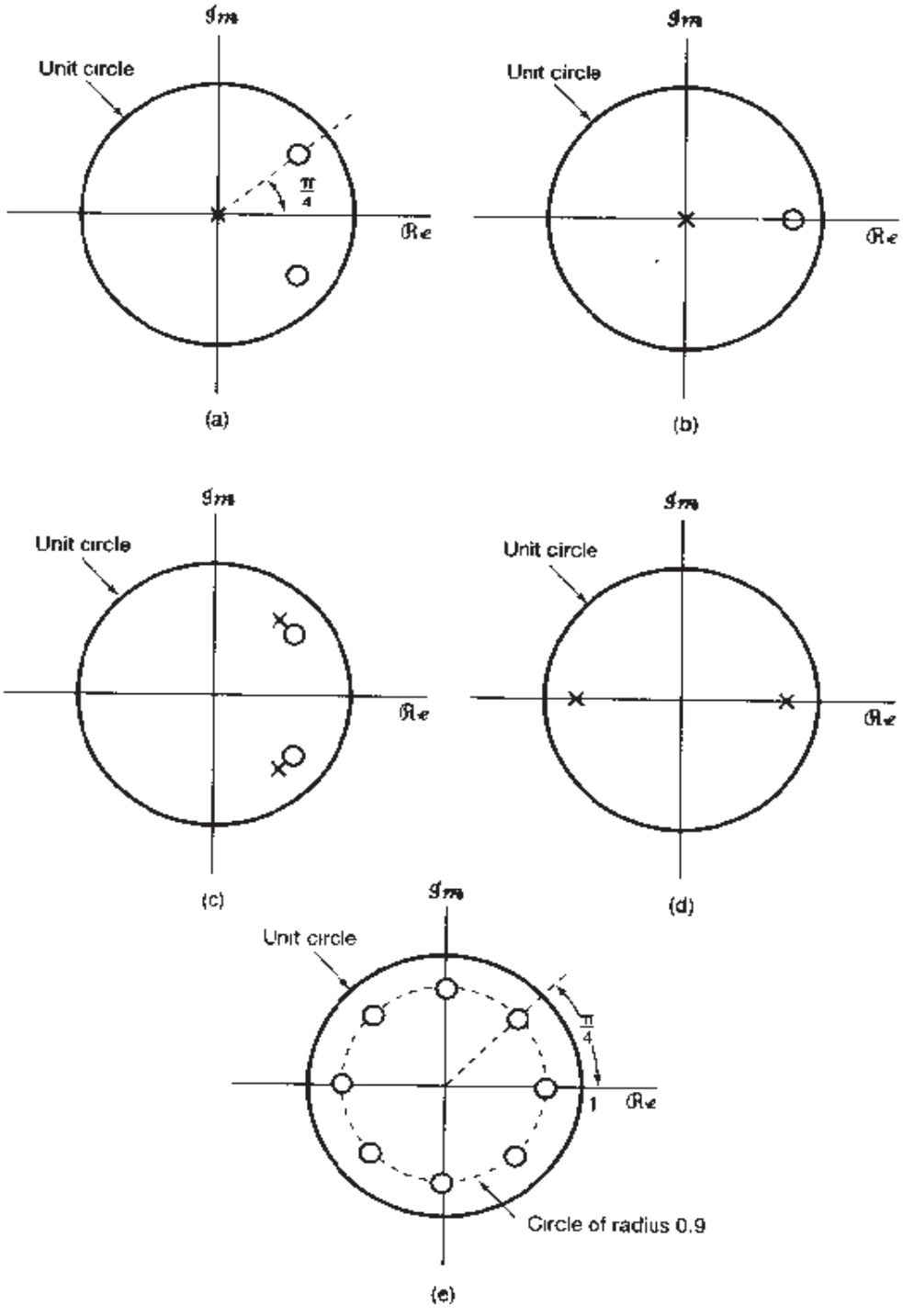



Figure P10.29

**10.30.** Consider a signal $y[n]$ which is related to two signals $x_1[n]$ and $x_2[n]$ by

$$y[n] = x_1[n+3] * x_2[-n+1]$$

where

$$x_1[n] = \left(\frac{1}{2}\right)^n u[n] \quad \text{and} \quad x_2[n] = \left(\frac{1}{3}\right)^n u[n].$$

Given that

$$a^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1-az^{-1}}, \quad |z| > |a|,$$

use properties of the $z$-transform to determine the $z$-transform $Y(z)$ of $y[n]$.

**10.31.** We are given the following five facts about a discrete-time signal $x[n]$ with $z$-transform $X(z)$:

1. $x[n]$ is real and right-sided.
2. $X(z)$ has exactly two poles.
3. $X(z)$ has two zeros at the origin.
4. $X(z)$ has a pole at $z = \frac{1}{2}e^{j\pi/3}$.
5. $X(1) = \frac{8}{3}$.

Determine $X(z)$ and specify its region of convergence.

**10.32.** Consider an LTI system with impulse response

$$h[n] = \begin{cases} a^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

and input

$$x[n] = \begin{cases} 1, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

**(a)** Determine the output $y[n]$ by explicitly evaluating the discrete convolution of $x[n]$ and $h[n]$.

**(b)** Determine the output $y[n]$ by computing the inverse $z$-transform of the product of the $z$-transforms of the input and the unit sample response.

**10.33.** **(a)** Determine the system function for the causal LTI system with difference equation

$$y[n] - \frac{1}{2}y[n-1] + \frac{1}{4}y[n-2] = x[n].$$

**(b)** Using $z$-transforms, determine $y[n]$ if

$$x[n] = \left(\frac{1}{2}\right)^n u[n].$$

**10.34.** A causal LTI system is described by the difference equation

$$y[n] = y[n-1] + y[n-2] + x[n-1].$$

**(a)** Find the system function $H(z) = Y(z)/X(z)$ for this system. Plot the poles and zeros of $H(z)$ and indicate the region of convergence.
**(b)** Find the unit sample response of the system.
**(c)** You should have found the system to be unstable. Find a stable (noncausal) unit sample response that satisfies the difference equation.

**10.35.** Consider an LTI system with input $x[n]$ and output $y[n]$ for which

$$y[n-1] - \frac{5}{2}y[n] + y[n+1] = x[n].$$

The system may or may not be stable or causal.

By considering the pole-zero pattern associated with the preceding difference equation, determine three possible choices for the unit sample response of the system. Show that each choice satisfies the difference equation.

**10.36.** Consider the linear, discrete-time, shift-invariant system with input $x[n]$ and output $y[n]$ for which

$$y[n-1] - \frac{10}{3}y[n] + y[n+1] = x[n].$$

The system is stable. Determine the unit sample response.

**10.37.** The input $x[n]$ and output $y[n]$ of a causal LTI system are related through the block-diagram representation shown in Figure P10.37.

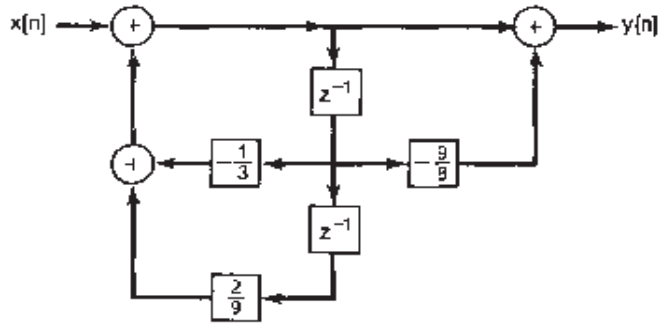



Figure P10.37

**(a)** Determine a difference equation relating $y[n]$ and $x[n]$.
**(b)** Is this system stable?

**10.38.** Consider a causal LTI system $S$ with input $x[n]$ and a system function specified as

$$H(z) = H_1(z)H_2(z),$$

where

$$H_1(z) = \frac{1}{1 + \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}$$

and

$$H_2(z) = 1 - \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}.$$

A block diagram corresponding to $H(z)$ may be obtained as a cascade connection of a block diagram for $H_1(z)$ followed by a block diagram for $H_2(z)$. The result is shown in Figure P10.38, in which we have also labeled the intermediate signals $e_1[n]$, $e_2[n]$, $f_1[n]$, and $f_2[n]$.

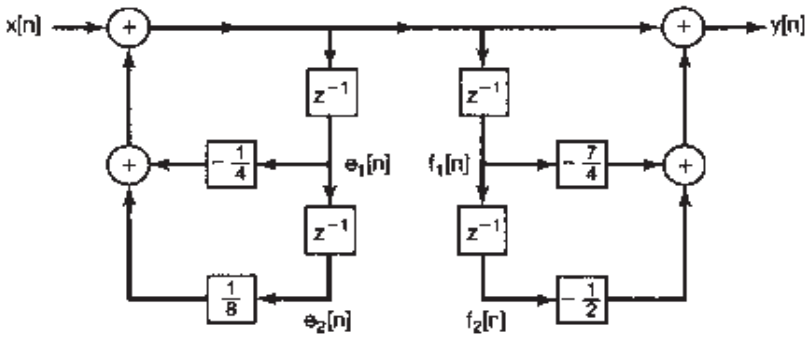



**Figure P10.38**

**(a)** How is $e_1[n]$ related to $f_1[n]$?
**(b)** How is $e_2[n]$ related to $f_2[n]$?
**(c)** Using your answers to the previous two parts as a guide, construct a direct-form block diagram for $S$ that contains only two delay elements.
**(d)** Draw a cascade-form block diagram representation for $S$ based on die observation that

$$H(z) = \left(\frac{1 + \frac{1}{4}z^{-1}}{1 + \frac{1}{2}z^{-1}}\right)\left(\frac{1 - 2z^{-1}}{1 - \frac{1}{4}z^{-1}}\right).$$

**(e)** Draw a parallel-form block diagram representation for $S$ based on the observation that

$$H(z) = 4 + \frac{5/3}{1 + \frac{1}{2}z^{-1}} - \frac{14/3}{1 - \frac{1}{4}z^{-1}}.$$

**10.39.** Consider the following three system functions corresponding to causal LTI systems:

$$H_1(z) = \frac{1}{(1 - z^{-1} + \frac{1}{4}z^{-2})(1 - \frac{2}{3}z^{-1} + \frac{1}{9}z^{-2})},$$

$$H_2(z) = \frac{1}{(1 - z^{-1} + \frac{1}{2}z^{-2})(1 - \frac{1}{2}z^{-1} + z^{-2})},$$

$$H_3(z) = \frac{1}{(1 - z^{-1} + \frac{1}{2}z^{-2})(1 - z^{-1} + \frac{1}{4}z^{-2})}.$$

(a) For each system function, draw a direct-form block diagram.
(b) For each system function, draw a block diagram that corresponds to the cascade connection of two second-order block diagrams. Each second-order block diagram should be in direct form.
(c) For each system function, determine whether there exists a block diagram representation which is the cascade of four first-order block diagrams with the constraint that all the coefficient multipliers must be real.

**10.40.** Determine the unilateral $z$-transform for each of the sequences in Problem 10.21

**10.41.** Consider the following two signals:

$$x_1[n] = \left(\frac{1}{2}\right)^{n+1} u[n+1],$$

$$x_2[n] = \left(\frac{1}{4}\right)^{n} u[n].$$

Let $\mathcal{X}_1(z)$ and $X_1(z)$ respectively be the unilateral and bilateral $z$-transforms of $x_1[n]$, and let $\mathcal{X}_2(z)$ and $X_2(z)$ respectively be the unilateral and bilateral $z$-transforms of $x_2[n]$.

(a) Take the inverse bilateral $z$-transform of $X_1(z)X_2(z)$ to determine $g[n] = x_1[n] * x_2[n]$.
(b) Take the inverse unilateral $z$-transform of $\mathcal{X}_1(z)\mathcal{X}_2(z)$ to obtain a signal $q[n]$ for $n \geq 0$. Observe that $q[n]$ and $g[n]$ are not identical for $n \geq 0$.

**10.42.** For each of the following difference equations and associated input and initial conditions, determine the zero-input and zero-state respenses by using the unilateral $z$-transform:

(a) $y[n] + 3y[n-1] = x[n]$,
$$x[n] = \left(\frac{1}{2}\right)^{n} u[n],$$
$$y[-1] = 1.$$

(b) $y[n] - \frac{1}{2}y[n-1] = x[n] - \frac{1}{2}x[n-1]$,
$$x[n] = u[n],$$
$$y[-1] = 0.$$

(c) $y[n] - \frac{1}{2}y[n-1] = x[n] - \frac{1}{2}x[n-1]$,
$$x[n] = u[n],$$
$$y[-1] = 1.$$

## ADVANCED PROBLEMS

**10.43.** Consider an even sequence $x[n]$ (i.e., $x[n] = x[-n]$) with rational $z$-transform $X(z)$.

**(a)** From the definition of the $z$-transform, show that

$$X(z) = X\left(\frac{1}{z}\right).$$

**(b)** From your results in part (a), show that if a pole (zero) of $X(z)$ occurs at $z = z_0$, then a pole (zero) must also occur at $z = 1/z_0$.

**(c)** Verify the result in part (b) for each of the following sequences:

**(1)** $\delta[n+1] + \delta[n-1]$

**(2)** $\delta[n+1] - \frac{5}{2}\delta[n] + \delta[n-1]$

**10.44.** Let $x[n]$ be a discrete-time signal with $z$-transform $X(z)$. For each of the following signals, determine the $z$-transform in terms of $X(z)$:

**(a)** $\Delta x[n]$, where $\Delta$ is the first-difference operator defined by

$$\Delta x[n] = x[n] - x[n-1]$$

**(b)** $x_1[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

**(c)** $x_1[n] = x[2n]$

**10.45.** Determine which of the following $z$-transforms could be the transfer function of a discrete-time linear system that is not necessarily stable, but for which the unit sample response is zero for $n < 0$. State your reasons clearly.

**(a)** $\dfrac{(1 - z^{-1})^2}{1 - \frac{1}{2}z^{-1}}$ **(b)** $\dfrac{(z-1)^2}{z - \frac{1}{2}}$

**(c)** $\dfrac{(z - \frac{1}{4})^5}{(z - \frac{1}{2})^6}$ **(d)** $\dfrac{(z - \frac{1}{4})^6}{(z - \frac{1}{2})^5}$

**10.46.** A sequence $x[n]$ is the output of an LTI system whose input is $s[n]$. The system is described by the difference equation

$$x[n] = s[n] - e^{8\alpha}s[n-8],$$

where $0 < \alpha < 1$.

**(a)** Find the system function

$$H_1(z) = \frac{X(z)}{S(z)},$$

and plot its poles and zeros in the $z$-plane. Indicate the region of convergence.

**(b)** We wish to recover $s[n]$ from $x[n]$ with an LTI system. Find the system function

$$H_2(z) = \frac{Y(z)}{X(z)}$$

such that $y[n] = s[n]$. Find all possible regions of convergence for $H_2(z)$, and for each, tell whether or not the system is causal or stable.

(c) Find all possible choices for the unit impulse response $h_2[n]$ such that

$$y[n] = h_2[n] * x[n] = s[n].$$

**10.47.** The following is known about a discrete-time LTI system with input $x[n]$ and output $y[n]$:

1. If $x[n] = (-2)^n$ for all $n$, then $y[n] = 0$ for all $n$
2. If $x[n] = (1/2)^n u[n]$ for all $n$, then $y[n]$ for all $n$ is of the form

$$y[n] = \delta[n] + a\left(\frac{1}{4}\right)^n u[n],$$

where $a$ is a constant.

(a) Determine the value of the constant $a$.

(b) Determine the response $y[n]$ if the input $x[n]$ is

$$x[n] = 1, \text{ for all } n.$$

**10.48.** Suppose a second-order causal LTI system has been designed with a real impulse response $h_1[n]$ and a rational system function $H_1(z)$. The pole-zero plot for $H_1(z)$ is shown in Figure P10.48(a). Now consider another causal second-order system with impulse response $h_2[n]$ and rational system function $H_2(z)$. The pole-zero plot for $H_2(z)$ is shown in Figure P10.48(b). Determine a sequence $g[n]$ such that the following three conditions hold:

(1) $h_2[n] = g[n]h_1[n]$

(2) $g[n] = 0$ for $n < 0$

(3) $\sum_{k=0}^{\infty} |g[k]| = 3$

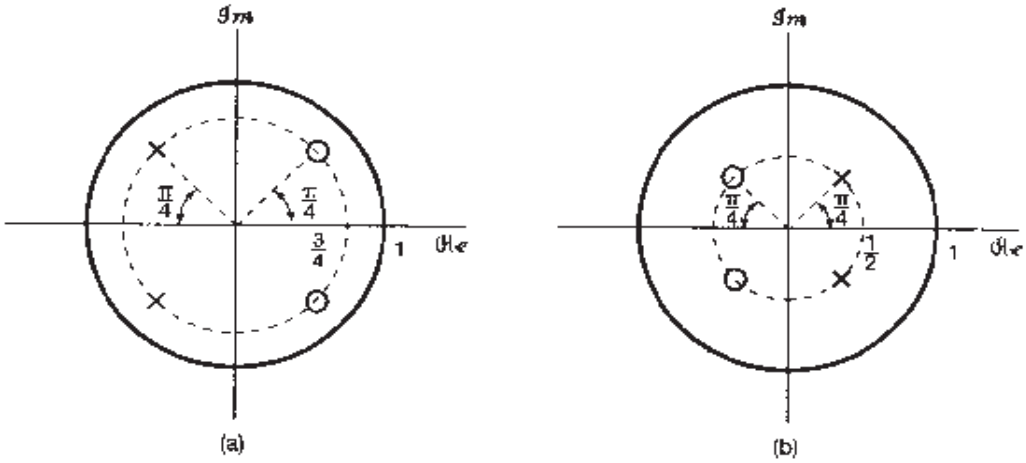



**Figure P10.48**

**10.49.** In Property 4 of Section 10.2, it was stated that if $x[n]$ is a right-sided sequence and if the circle $|z| = r_0$ is in the ROC, then all finite values of $z$ for which $|z| < r_0$ will also be in the ROC. In this discussion an intuitive explanation was given. A more formal argument parallels closely that used for Property 4 of Section 9.2, relating to the Laplace transform. Specifically, consider a right-sided sequence

$$x[n] = 0, \quad n < N_1,$$

and for which

$$\sum_{n=-\infty}^{\infty} |x[n]| r_0^{\,n} = \sum_{n=N_1}^{\infty} |x[n]| r_0^{-n} < \infty.$$

Then if $r_0 \leq r_1$,

$$\sum_{n=N_1}^{\infty} |x[n]| r_1^{\,n} \leq A \sum_{n=N_1}^{\infty} |x[n]| r_0^{-n}, \tag{P10.49-1}$$

where $A$ is a positive constant.

(a) Show that eq (P10.49–1) is true, and determine the constant $A$ in terms of $r_0$, $r_1$, and $N_1$.

(b) From your result in part (a), show that Property 4 of Section 10.2 follows.

(c) Develop an argument similar to the foregoing one to demonstrate the validity of Property 5 of Section 10.2.

**10.50.** A discrete-time system with the pole-zero pattern shown in Figure P10.50(a) is referred to as a first-order all-pass system, since the magnitude of the frequency response is constant regardless of frequency.

(a) Demonstrate algebraically that $|H(e^{j\omega})|$ is constant.

To demonstrate the same property geometrically, consider the vector diagram in Figure P10.50(b). We wish to show that the length of $v_2$ is proportional to the length of $v_1$ independently of the frequency $\omega$.

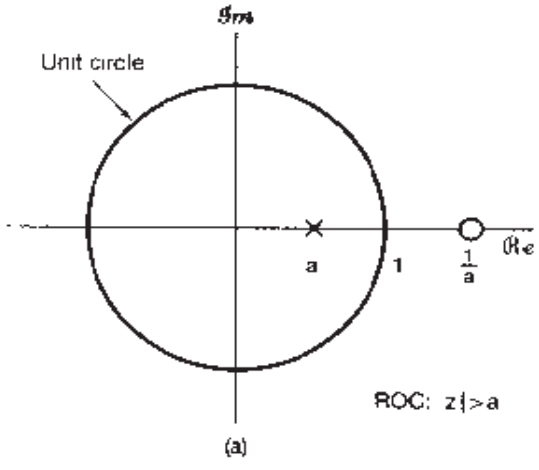



Figure P10.50a

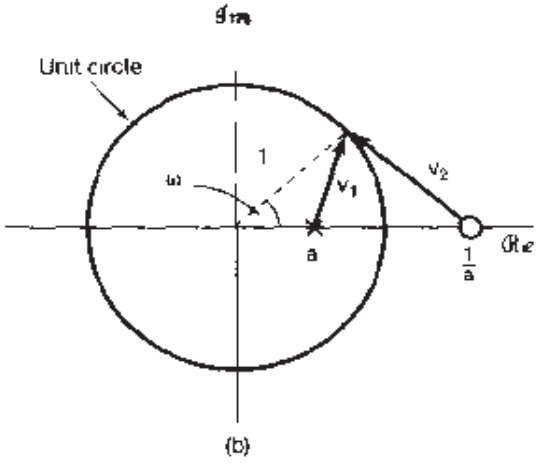



**Figure P10.50b**

**(b)** Express the length of $v_1$ using the law of cosines and the fact that $v_1$ is one leg of a triangle for which the other two legs are the unit vector and a vector of length $a$.

**(c)** In a manner similar to that in part (b), determine the length of $v_2$ and show that it is proportional in length to $v_1$ independently of $\omega$.

**10.51.** Consider a real-valued sequence $x[n]$ with rational $z$-transform $X(z)$.

**(a)** From the definition of the $z$-transform, show that

$$X(z) = X^*(z^*).$$

**(b)** From your result in part (a), show that if a pole (zero) of $X(z)$ occurs at $z = z_0$, then a pole (zero) must also occur at $z = z_0^*$.

**(c)** Verify the result in part (b) for each of the following sequences:

**(1)** $x[n] = (\frac{1}{2})^n u[n]$

**(2)** $x[n] = \delta[n] - \frac{1}{2}\delta[n-1] + \frac{1}{4}\delta[n-2]$

**(d)** By combining your results in part (b) with the result of Problem 10.43(b), show that for a real, even sequence, if there is a pole (zero) of $H(z)$ at $z = \rho e^{j\theta}$, then there is also a pole (zero) of $H(z)$ at $z = (1/\rho)e^{j\theta}$ and at $z = (1/\rho)e^{-j\theta}$.

**10.52.** Consider a sequence $x_1[n]$ with $z$-transform $X_1(z)$ and a sequence $x_2[n]$ with $z$-transform $X_2(z)$, where

$$x_2[n] = x_1[-n].$$

Show that $X_2(z) = X_1(1/z)$, and from this, show that if $X_1(z)$ has a pole (or zero) at $z = z_0$, then $X_2(z)$ has a pole (or zero) at $z = 1/z_0$.

**10.53.** **(a)** Carry out the proof for each of the following properties in Table 10.1.

**(1)** Property set forth in Section 10.5.2

**(2)** Property set forth in Section 10.5.3

**(3)** Property set forth in Section 10.5.4

(b) With $X(z)$ denoting the $z$-transform of $x[n]$ and $R_x$ the ROC of $X(z)$, determine, in terms of $X(z)$ and $R_x$, the $z$-transform and associated ROC for each of the following sequences:
(1) $x^*[n]$
(2) $z_0^n x[n]$, where $z_0$ is a complex number

**10.54.** In Section 10.5.9, we stated and proved the initial-value theorem for causal sequences.
(a) State and prove the corresponding theorem if $x[n]$ is anticausal (i.e., if $x[n] = 0, n > 0$).
(b) Show that if $x[n] = 0, n < 0$, then

$$x[1] = \lim_{z \to \infty} z(X(z) - x[0]).$$

**10.55.** Let $x[n]$ denote a causal sequence (i.e., if $x[n] = 0, n < 0$) for which $x[0]$ is nonzero and finite.
(a) Using the initial-value theorem, show that there are no poles or zeros of $X(z)$ at $z = \infty$.
(b) Show that, as a consequence of your result in part (a), the number of poles of $X(z)$ in the finite $z$-plane equals the number of zeros of $X(z)$ in the finite $z$-plane. (The finite $z$-plane excludes $z = \infty$.)

**10.56.** In Section 10.5.7, we stated the convolution property for the $z$-transform. To show that this property holds, we begin with the convolution sum expressed as

$$x_3[n] = x_1[n] * x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]. \qquad \text{(P10.56–1)}$$

(a) By taking the $z$-transform of eq. (P10.56–1) and using eq. (10.3), show that

$$X_3(z) = \sum_{k=-\infty}^{\infty} x_1[k]\hat{X}_2(z),$$

where $\hat{X}_2(z)$ is the transform of $x_2[n-k]$.
(b) Using your result in part (a) and property 10.5.2 in Table 10.1, show that

$$X_3(z) = X_2(z) \sum_{k=-\infty}^{\infty} x_1[k]z^{-k}.$$

(c) From part (b), show that

$$X_3(z) = X_1(z)X_2(z)$$

as stated in eq. (10.81).

**10.57.** Let

$$X_1(z) = x_1[0] + x_1[1]z^{-1} + \cdots + x_1[N_1]z^{-N_1},$$
$$X_2(z) = x_2[0] + x_2[1]z^{-1} + \cdots + x_2[N_2]z^{-N_2}.$$

Define

$$Y(z) = X_1(z)X_2(z).$$

and let

$$Y(z) = \sum_{k=0}^{M} y[k]z^{-k}.$$

(a) Express $M$ in terms of $N_1$ and $N_2$.
(b) Use polynomial multiplication to determine $y[0]$, $y[1]$, and $y[2]$.
(c) Use polynomial multiplication to show that, for $0 \leq k \leq M$,

$$y[k] = \sum_{m=-\infty}^{\infty} x_1[m]x_2[k-m].$$

**10.58.** A minimum-phase system is a system that is causal and stable and for which the inverse system is also causal and stable. Determine the necessary constraints on the location in the $z$-plane of the poles and zeros of the system function of a minimum-phase system.

**10.59.** Consider the digital filter structure shown in Figure P10.59.

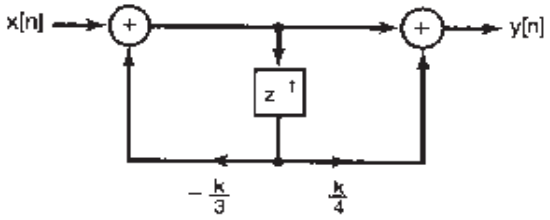



Figure P10.59

(a) Find $H(z)$ for this causal filter. Plot the pole-zero pattern and indicate the region of convergence.
(b) For what values of the $k$ is the system stable?
(c) Determine $y[n]$ if $k = 1$ and $x[n] = (2/3)^n$ for all $n$.

**10.60.** Consider a signal $x[n]$ whose unilateral $z$-transform is $\mathcal{X}(z)$. Show that the unilateral $z$-transform of $y[n] = x[n+1]$ may be specified as

$$\mathcal{Y}(z) = z\mathcal{X}(z) - zx[0].$$

**10.61.** If $\mathcal{X}(z)$ denotes the unilateral $z$-transform of $x[n]$, determine, in terms of $\mathcal{X}(z)$, the unilateral $z$-transform of:
(a) $x[n+3]$ (b) $x[n-3]$ (c) $\sum_{k=-\infty}^{n} x[k]$

## EXTENSION PROBLEMS

**10.62.** The autocorrelation sequence of a sequence $x[n]$ is defined as

$$\phi_{xx}[n] = \sum_{k=-\infty}^{\infty} x[k]x[n+k].$$

Determine the $z$-transform of $\phi_{xx}[n]$ in terms of the $z$-transform of $x[n]$

**10.63.** By using the power-series expansion

$$\log(1 - w) = -\sum_{i=1}^{\infty} \frac{w^i}{i}, \quad |w| < 1,$$

determine the inverse of each of the following two $z$-transforms:

**(a)** $X(z) = \log(1 - 2z), \ |z| < \frac{1}{2}$

**(b)** $X(z) = \log(1 - \frac{1}{2}z^{-1}), \ |z| > \frac{1}{2}$

**10.64.** By first differentiating $X(z)$ and using the appropriate properties of the $z$-transform, determine the sequence for which the $z$-transform is each of the following:

**(a)** $X(z) = \log(1 - 2z), \ |z| < \frac{1}{2}$

**(b)** $X(z) = \log(1 - \frac{1}{2}z^{-1}), \ |z| > \frac{1}{2}$

Compare your results for (a) and (b) with the results obtained in Problem 10.63, in which the power-series expansion was used.

**10.65.** The *bilinear transformation* is a mapping for obtaining a rational $z$-transform $H_d(z)$ from a rational Laplace transform $H_c(s)$. This mapping has two important properties:

1. If $H_c(s)$ is the Laplace transform of a causal and stable LTI system, then $H_d(z)$ is the $z$-transform of a causal and stable LTI system.
2. Certain important characteristics of $|H_c(j\omega)|$ are preserved in $|H_d(e^{j\omega})|$.

In this problem, we illustrate the second of these properties for the case of all-pass filters.

**(a)** Let

$$H_c(s) = \frac{a - s}{s + a},$$

where $a$ is real and positive. Show that

$$|H_c(j\omega)| = 1.$$

**(b)** Let us now apply the bilinear transformation to $H_c(s)$ in order to obtain $H_d(z)$. That is,

$$H_d(z) = H_c(s)\Big|_{s = \frac{1 - z^{-1}}{1 + z^{-1}}}.$$

Show that $H_d(z)$ has one pole (which is inside the unit circle) and one zero (which is outside the unit circle).

**(c)** For the system function $H_d(z)$ derived in part (c), show that $|H_d(e^{j\omega})| = 1$.

**10.66.** The bilinear transformation, introduced in the previous problem, may also be used to obtain a discrete-time filter, the magnitude of whose frequency response is similar to the magnitude of the frequency response of a given continuous-time low-pass filter. In this problem, we illustrate the similarity through the example of a continuous-time second-order Butterworth filter with system function $H_c(s)$.

(a) Let

$$H_d(z) = H_c(s)\Big|_{s=\frac{1-z^{-1}}{1+z^{-1}}}.$$

Show that

$$H_d(e^{j\omega}) = H_c\left(j\tan\frac{\omega}{2}\right).$$

(b) Given that

$$H_c(s) = \frac{1}{(s + e^{j\pi/4})(s + e^{-j\pi/4})}$$

and that the corresponding filter is causal, verify that $H_c(0) = 1$, that $|H_c(j\omega)|$ decreases monotonically with increasing positive values of $\omega$, that $|H_c(j)|^2 = 1/2$ (i.e., that $\omega_c = 1$ is the half-power frequency), and that $H_c(\infty) = 0$.

(c) Show that if the bilinear transformation is applied to $H_c(s)$ of part (b) in order to obtain $H_d(z)$, then the following may be asserted about $H_d(z)$ and $H_d(e^{j\omega})$:

1. $H_d(z)$ has only two poles, both of which are *inside* the unit circle.
2. $H_d(e^{j0}) = 1$.
3. $|H_d(e^{j\omega})|$ decreases monotonically as $\omega$ goes from 0 to $\pi$.
4. The half-power frequency of $H_d(e^{j\omega})$ is $\pi/2$.